

Verification & Validation Protocol

Document ID	VVP-MEC-001
Device	Six-Sense Diagnostic Microneedle Patch
Testing Phase	Design Verification (Mechanical Integrity)
Regulatory Alignment	FDA 21 CFR Part 820.30, ISO 13485, 21 CFR Part 11

1. Objective

To verify that the structural integrity of the Six-Sense microneedle array can withstand standard application forces without fracturing or undergoing critical deformation, adhering to FDA Design Control requirements (21 CFR Part 820.30).

2. Scope

This protocol applies to the simulated compressive load testing of 50 randomly sampled microneedle arrays across two manufacturing runs (Lot A and Lot B). This test evaluates the physical yield limit and structural integrity of the individual microneedles under direct axial load.

3. Equipment & Methodology

In accordance with Good Laboratory Practices (GLP) and biomechanical testing standards:

- **Equipment:** Simulated Instron Universal Testing Machine (Load Cell).
- **Methodology:** A continuous downward compressive force is applied to the central axis of the patch at a constant displacement rate until physical failure.
- **Data Capture:** The exact peak force (Newtons) and displacement (millimeters) at the precise moment of structural fracture are recorded and exported to a CSV matrix.

4. Acceptance Criteria

Minimum Yield Stress Threshold: 120.0 MPa
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To pass batch release, the calculated engineering stress (Force divided by Cross-Sectional Area) at the moment of fracture must be ≥ 120.0 MPa.

Devices fracturing below this threshold fail inspection. If a specific manufacturing lot demonstrates a high failure rate, it must be flagged for mechanical root-cause analysis and Corrective and Preventive Action (CAPA).

5. Automated Analysis & Data Integrity

To comply with FDA 21 CFR Part 11 (Electronic Records and Signatures):

- Raw load-cell data is processed through an automated Python QA pipeline (Pandas/NumPy) to calculate true engineering stress and strain, eliminating manual transcription errors.
- The script automatically tags failing units against the acceptance criteria and generates a stress-strain scatter plot to provide a locked visual regulatory audit of batch consistency.